



Advanced Traffic (xx-xxx)
Class Project Part 4

Part 4: Simulation Engine & Data Logging

Goal:

1. `simpy`
2. `sympy`
3. Read demand file and generate vehicles properly.
4. Build the simulation loop and collect data.

Tasks

1. Implement a simulation engine (using `simpy` or a custom loop).
2. Generate vehicles from demand data and move them according to logic.
3. Log vehicle positions and states to a `DataFrame`.

Output

1. Implement Engine file to run the simulation.
2. Modified files of python to integrate with Engine.py file.
3. Brief report about the code(documentation in all files with a `read_me.md` file.)
4. Presenting the code for me and answer the questions.
5. Present the Log file(as CSV)

Time

Week 3-4

Grade Weight

40%(Here you get all the grades from part 2 and 3 combined with 4. [2: 20, 3:10, 4:10].) So if I see problem in the vehicles and network, you will lose a portion of 20 %. If the problem exists in traffic light, you gave portion of 10 percent. if the problem exists on the Engine part, You most probably lose a portion of 10 percent with a portion of 30 because the code will not run and we can not make sure all the rest is ok or not.